

Load-Serving Entity (LSE) / Utility		Gather Input & Prepare Collect demand, contracts, and market data		Where AI Helps
Load-Serving Entity (LSE) / Utility (Focused on serving load reliably at least cost over short and long horizons) Grid Operator / Dispatcher Short-Term Trader / Scheduler Load Forecaster Market Analyst Resource Planner (IRP / Capacity) Portfolio Manager Risk Manager Financial / Investment Analyst	Real Time Load vs Forecast Deviation Generation & Outage Status Real-Time Price & Congestion Signals Reserve / Ancillary Status Renewable Variability & Weather Impact System Imbalance / Shortage Risk Load minus renewable generation Critical for real-time balancing	Goal	Ensure reliable real-time power supply Minimize imbalance costs and price exposure Maintain grid stability under uncertainty	Data integration & cleaning
		Actions	Monitor real-time load, generation, outages, and congestion Continuously compare forecast vs actual and trigger adjustments Adjust dispatch or procure power as needed	
		Thoughts	“Is the forecast still accurate?” “Do we have enough capacity right now?” “Are prices or outages creating risk?”	
		Pain Point	Inaccurate load or renewable forecasts Sudden outages and unexpected events Limited real-time visibility and coordination	
		Emotions	High alert during volatile conditions Stress from uncertainty and rapid decisions Relief when system operates smoothly	
		Tool	PLEXOS (short-term simulation) SCADA / EMS systems Market & weather data dashboards + ISO Market Systems (RTM portals)	
	Day Ahead / Forward Demand & Forecast Reports Price & Market Simulation Reports Resource Planning (IRP / Capacity) Procurement & Bidding Report Reliability & Adequacy Report Financial / Investment Reports	Goal	Procure lowest-cost and reliable energy Optimize portfolio (generation, contracts, market) Ensure adequacy and regulatory compliance	Example AI Capability Auto-ingest data from ISO, weather, fuel APIs + anomaly detection
		Actions	Gather demand, weather, fuel, and asset data Run PLEXOS simulations and scenario analysis Validate results and prepare DAM bids	
		Thoughts	“Is my forecast accurate enough?” “What if key assumptions change (renewables, demand)?” “Which scenario is most reliable for decisions?”	
		Pain Point	Data dependency across multiple teams High model complexity and uncertainty Tight timelines and validation challenges	
		Emotions	Focused during analysis Pressure near deadlines Confidence when results align with expectations	
		Tool	PLEXOS (modelling & simulation) Excel / data pipelines Forecasting & market data tools + ISO Market Systems (DAM portals)	

Load-Serving Entity (LSE) / Utility		Simulation Setting Configure procurement and demand models		Where AI Helps
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		Actions	Select real-time horizon, step size, and update frequency for the run Latest inputs: load, outages, fuel availability, renewable forecast, reserves, congestion constraints Choose required outputs such as price, dispatch, reserve, imbalance, and shortage reports	
		Thoughts	“Did I set the right horizon and granularity for this operating window?” “Are all critical constraints and latest system conditions included?” “Will the outputs answer the operational decision I need right now?”	
		Pain Point	Wrong horizon or step size can make results less useful for operations Missing or stale inputs can distort dispatch and balancing outcomes Too many report options make it hard to focus on the most decision-relevant outputs	
		Emotions	Careful while configuring high-impact short-term runs Tense when time is limited and system conditions are changing Reassured when settings match the real operating situation	
		Tool	PLEXOS short-term market / dispatch simulation SCADA / EMS and ISO real-time market inputs Market, outage, weather, and renewable forecast data sources	
	Day Ahead / Forward Demand & Forecast Reports Price & Market Simulation Reports Resource Planning (IRP / Capacity) Procurement & Bidding Report Reliability & Adequacy Report Financial / Investment Reports	Goal	Set up simulations for day-ahead, week-ahead, or forward planning decisions Align horizon, assumptions, and scenarios with procurement, planning, and compliance needs Produce reports for bids, adequacy, cost, and portfolio decisions	Example AI Capability AI suggests parameters (load growth, reserves, constraints)
		Actions	Define study horizon, scenario set, and market/planning assumptions Configure model settings for dispatch, capacity, reliability, fuel, and policy constraints Select output reports such as forecast, unit commitment, adequacy, bidding, and investment reports	
		Thoughts	“Is this the right horizon for the business question: day-ahead, week-ahead, or longer?” “Do my simulation assumptions reflect realistic demand, fuel, and renewable conditions?” “Will these reports be strong enough to support bids, planning, or stakeholder review?”	
		Pain Point	Choosing the right horizon and scenario scope is not always straightforward Model settings become complex when many constraints and uncertainties are included Results must be defensible against benchmark, history, and stakeholder scrutiny	
		Emotions	Focused when structuring the simulation correctly Uncertain when comparing multiple scenario setups Confident when model design is aligned with the decision context	
		Tool	PLEXOS modelling and scenario configuration Excel / data pipelines for structured inputs Forecast, fuel, market, and policy data sources plus ISO DAM portals	

Load-Serving Entity (LSE) / Utility		Execute Back cast Model & RUN Validate model vs historical demand & cost		Where AI Helps
Load-Serving Entity (LSE) / Utility (Focused on serving load reliably at least cost over short and long horizons) Grid Operator / Dispatcher Short-Term Trader / Scheduler Load Forecaster Market Analyst Resource Planner (IRP / Capacity) Portfolio Manager Risk Manager Financial / Investment Analyst	Real Time Load vs Forecast Deviation Generation & Outage Status Real-Time Price & Congestion Signals Reserve / Ancillary Status Renewable Variability & Weather Impact System Imbalance / Shortage Risk Load minus renewable generation Critical for real-time balancing	Goal	Recreate actual short-term operating conditions using historical data Test whether dispatch, prices, and imbalance behaviour match what actually happened Improve confidence in real-time forecasting and operating logic	Error detection & calibration Example AI Capability AI highlights mismatch (forecast vs actual) and suggests corrections
		Actions	Load historical load, outage, renewable, weather, and RT market data Run the back cast for selected historical hours or operating periods Compare simulated results with actual dispatch, price, and shortage outcomes	
		Thoughts	“Does the model reproduce what actually happened?” “Which hours or events are creating the biggest error?” “Can this setup be trusted for live operational decisions?”	
		Pain Point	Historical operational data may be incomplete or inconsistent Volatile events are hard to recreate accurately It is difficult to isolate whether errors come from data, assumptions, or model setup	
		Emotions	Focused while replaying critical periods Frustrated when results do not match history More confident when back cast results align with actual outcomes	
		Tool	PLEXOS short-term simulation / back cast setup Historical SCADA / EMS / ISO real-time data Weather, outage, and renewable history datasets	
		Goal	Validate the base model against historical market outcomes before using it forward Calibrate assumptions for demand, fuel, renewables, and constraints Build a defensible benchmark for future scenarios, bids, and planning	
		Actions	Run back cast using historical demand, weather, fuel, outage, and market data Compare benchmark actual market results with base model outputs Adjust assumptions and rerun until the model tracks history closely	
	Day Ahead / Forward Demand & Forecast Reports Price & Market Simulation Reports Resource Planning (IRP / Capacity) Procurement & Bidding Report Reliability & Adequacy Report Financial / Investment Reports	Thoughts	“Is the base model close enough to the benchmark?” “Which assumption is causing the gap between simulated and actual results?” “Is this accurate enough to support scenarios, bids, or stakeholder review?”	
		Pain Point	Limited historical data can weaken back cast accuracy Binding constraints and real market behaviour are hard to capture correctly Results must stand up to benchmarking and stakeholder scrutiny	
		Emotions	Analytical when comparing benchmark vs model Uncertain when large deviations appear Reassured when results are defensible and close to history	
		Tool	PLEXOS base model / benchmark setup Historical ISO market, fuel, weather, and load datasets Excel / data pipelines for calibration and comparison	

Load-Serving Entity (LSE) / Utility		Analyse Solution Analyse cost, supply mix, and adequacy		Where AI Helps
Load-Serving Entity (LSE) / Utility (Focused on serving load reliably at least cost over short and long horizons) Grid Operator / Dispatcher Short-Term Trader / Scheduler Load Forecaster Market Analyst Resource Planner (IRP / Capacity) Portfolio Manager Risk Manager Financial / Investment Analyst	Real Time Load vs Forecast Deviation Generation & Outage Status Real-Time Price & Congestion Signals Reserve / Ancillary Status Renewable Variability & Weather Impact System Imbalance / Shortage Risk Load minus renewable generation Critical for real-time balancing	Goal	Assess whether simulated results support real-time operational decisions Identify deviations in price, dispatch, congestion, and imbalance Ensure reliability and cost outcomes are within acceptable limits	Insight generation
		Actions	Review outputs: price (LMP), dispatch, reserves, congestion, imbalance Compare simulation vs current/actual system conditions Flag anomalies and trigger corrective operational actions	
		Thoughts	“Do these results reflect current system reality?” “Are there any risks (price spikes, shortages, congestion)?” “Do I need to adjust dispatch or procure more power?”	
		Pain Point	Too many outputs make it hard to identify key issues quickly Real-time deviations can be difficult to interpret under time pressure Lack of clear linkage between results and operational decisions	
		Emotions	Alert while scanning for risks and anomalies Pressure to interpret results quickly Relief when results confirm stable system operation	
		Tool	PLEXOS output dashboards (short-term results) SCADA / EMS visualization tools Real-time market and congestion dashboards	
	Day Ahead / Forward Demand & Forecast Reports Price & Market Simulation Reports Resource Planning (IRP / Capacity) Procurement & Bidding Report Reliability & Adequacy Report Financial / Investment Reports	Goal	Evaluate simulation outputs for decision-making (bidding, planning, investment) Compare scenarios to identify optimal strategy Validate results against benchmark, constraints, and business objectives	Example AI Capability “Top 3 cost drivers this month: fuel + peak demand + congestion”
		Actions	Analyse outputs: price forecasts, unit commitment, generation mix, adequacy Compare multiple scenarios (demand, renewables, fuel, policy changes) Select best scenario for bidding, procurement, or planning decisions	
		Thoughts	“Which scenario gives the best cost vs reliability outcome?” “Are results consistent with expectations and historical trends?” “Can I defend this analysis to stakeholders or regulators?”	
		Pain Point	Difficulty comparing multiple scenarios effectively High effort required to validate and interpret outputs Need to justify results under stakeholder/regulatory scrutiny	
		Emotions	Analytical when reviewing outputs Uncertain when scenarios give conflicting results Confident when a clear optimal solution emerges	
		Tool	PLEXOS output reports & scenario comparison tools Excel / BI dashboards for analysis Visualization tools for trends, cost, and reliability metrics	

Load-Serving Entity (LSE) / Utility		Forecast/Base Model & Run Forecast demand and procurement needs		Where AI Helps
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		Actions	Run short-term forecast models using latest load, weather, and outage inputs Update base model with current system conditions and constraints Generate forward-looking outputs (price, dispatch, imbalance risk)	
		Thoughts	“How accurate are these short-term predictions?” “Are there any upcoming risks we should act on early?” “Do we need to adjust positions before real-time?”	
		Pain Point	Forecast accuracy is sensitive to weather and renewable variability Rapid system changes can make predictions outdated quickly Limited visibility into sudden outages or market behaviour	
		Emotions	Alert while anticipating near-term risks Uncertain due to forecast variability Confident when predictions align with trends	
		Tool	PLEXOS short-term forecasting / dispatch models Weather & renewable forecast tools Market data feeds and RTM signals	
		Goal	Build future scenarios for demand, price, and system behaviour Generate base model outputs for bidding, planning, and investment Support forward decisions with reliable and defensible forecasts	Example AI Capability AI generates load forecast using weather + history
		Actions	Run base model using forecasted demand, fuel, weather, and policy inputs Generate multiple scenarios (high/low demand, renewable variability, fuel changes) Produce outputs for price, unit commitment, adequacy, and cost	
	Day Ahead / Forward Demand & Forecast Reports Price & Market Simulation Reports Resource Planning (IRP / Capacity) Procurement & Bidding Report Reliability & Adequacy Report Financial / Investment Reports	Thoughts	“Which forecast scenario is most realistic?” “How sensitive are results to key assumptions?” “Can I rely on this for bidding or planning decisions?”	
		Pain Point	High uncertainty in long-term assumptions (demand, fuel, renewables) Managing multiple scenarios becomes complex Ensuring forecast credibility for stakeholders and regulators	
		Emotions	Analytical while exploring scenarios Uncertain when results vary across assumptions Confident when forecasts are consistent and validated	
		Tool	PLEXOS forecasting & scenario modelling Forecasting tools (load, price, renewables, fuel) Data pipelines, market data, and ISO DAM systems	

Forecast/Base Model + Scenario-A,,, & Run | Test supply/demand & investment scenarios **Where AI Helps**

Load-Serving Entity (LSE) / Utility (Focused on serving load reliably at least cost over short and long horizons) Grid Operator / Dispatcher Short-Term Trader / Scheduler Load Forecaster Market Analyst Resource Planner (IRP / Capacity) Portfolio Manager Risk Manager	Real Time	Goal	Test multiple short-term scenarios (load spikes, outages, renewable swings) Identify best operational response under uncertainty Reduce risk before events occur in real-time operations
	Load vs Forecast Deviation	Actions	Create scenarios (high load, outage event, low renewable output) Run simulations for each scenario using updated system conditions Compare outputs (price, dispatch, imbalance) across scenarios
	Generation & Outage Status		
	Real-Time Price & Congestion Signals		
	Reserve / Ancillary Status	Thoughts	“What happens if conditions worsen suddenly?” “Which scenario shows the highest operational risk?” “Do we need a contingency plan before real-time?”
	Renewable Variability & Weather Impact		
	System Imbalance / Shortage Risk		
	Load minus renewable generation	Pain Point	Scenario setup can be time-consuming under tight timelines Hard to prioritize which scenarios matter most Results may vary significantly across scenarios
	Critical for real-time balancing		
	Emotions	Alert while preparing for uncertainty Pressure to evaluate multiple outcomes quickly Confidence when a clear contingency plan is identified	
	Tool	PLEXOS scenario simulation (short-term runs) Real-time data feeds (load, outages, weather) Market dashboards for scenario comparison	
Financial / Investment Analyst Demand & Forecast Reports Price & Market Simulation Reports Resource Planning (IRP / Capacity) Procurement & Bidding Report Reliability & Adequacy Report Financial / Investment Reports	Day Ahead / Forward	Goal	Evaluate multiple future scenarios for planning, bidding, and investment Understand sensitivity of results to key drivers (demand, fuel, renewables) Select optimal strategy under uncertainty
		Actions	Define scenarios (high/low demand, fuel price changes, renewable variability, policy shifts) Run PLEXOS simulations for each scenario using base model Compare outputs (price, unit commitment, adequacy, cost) to select best scenario
		Thoughts	“Which scenario is most realistic vs worst-case?” “How sensitive are results to key assumptions?” “Which scenario should drive our bidding or planning decision?”
		Pain Point	Managing large number of scenarios becomes complex Difficulty comparing outputs across scenarios effectively Need to justify scenario selection to stakeholders
	Emotions	Analytical when evaluating multiple scenarios Uncertain when scenarios give conflicting signals Confident when one scenario clearly stands out	
	Tool	PLEXOS scenario management & simulation Excel / BI tools for comparison Forecasting, fuel, and policy data sources + ISO DAM systems	

Scenario generation

Example AI Capability

AI creates scenarios
(high demand, low
renewable, price
spike)

Load-Serving Entity (LSE) / Utility		Compare Scenarios Select least-cost reliable option		Where AI Helps
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		Actions	Compare scenario outputs (price, dispatch, reserves, imbalance) Identify worst-case vs best-case scenarios Choose and trigger the most reliable contingency plan	
		Thoughts	“Which scenario is safest to act on right now?” “Are we covered if conditions worsen?” “Do we need immediate corrective action?”	
		Pain Point	Time pressure limits deep comparison Hard to interpret multiple scenario differences quickly Risk of missing critical edge-case scenarios	
		Emotions	High pressure during rapid decision-making Alert when risk scenarios emerge Confidence when a clear action path is identified	
		Tool	PLEXOS scenario comparison outputs Real-time dashboards (price, congestion, imbalance) SCADA / EMS operational systems	
		Goal	Select optimal scenario for bidding, planning, or investment decisions Balance cost, reliability, and risk across scenarios Ensure decisions are defensible and aligned with strategy	Example AI Capability AI recommends best scenario: “Lowest cost with acceptable risk”
		Actions	Compare scenario outputs (price, unit commitment, adequacy, cost) Evaluate trade-offs across demand, fuel, and renewable assumptions Select final scenario for DAM bids or planning decisions	
	Day Ahead / Forward Demand & Forecast Reports Price & Market Simulation Reports Resource Planning (IRP / Capacity) Procurement & Bidding Report Reliability & Adequacy Report Financial / Investment Reports	Thoughts	“Which scenario gives best cost vs reliability outcome?” “How sensitive is the result to key assumptions?” “Can I defend this choice to stakeholders/regulators?”	
		Pain Point	Too many scenarios to compare efficiently Conflicting results across scenarios Need strong justification for final selection	
		Emotions	Analytical during comparison Uncertain when no clear winner exists Confident when optimal scenario is clear	
		Tool	PLEXOS scenario comparison & reporting Excel / BI dashboards Market, fuel, and forecast data sources	

Load-Serving Entity (LSE) / Utility		Results Procure power and serve customers		Where AI Helps
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		Actions	Review final outputs (price, dispatch, reserves, imbalance) Execute operational decisions (dispatch changes, procure power) Monitor outcomes and adjust in real time	
		Thoughts	“Are we operating within safe and cost limits?” “Did the decision perform as expected?” “Do we need further adjustments?”	
		Pain Point	Continuous monitoring required as conditions change Limited time to react to unexpected deviations Difficulty linking results directly to prior assumptions	
		Emotions	Relief when system runs smoothly Stress during unexpected deviations Confidence when outcomes match expectations	
		Tool	SCADA / EMS operational systems Real-time market dashboards (LMP, congestion) PLEXOS output monitoring (short-term)	
	Day Ahead / Forward Demand & Forecast Reports Price & Market Simulation Reports Resource Planning (IRP / Capacity) Procurement & Bidding Report Reliability & Adequacy Report Financial / Investment Reports	Goal	Deliver final outputs for bidding, planning, and investment decisions Ensure results meet cost, reliability, and regulatory objectives Provide defensible insights for stakeholders	
		Actions	Finalize outputs (price forecast, unit commitment, adequacy, cost) Submit DAM bids or planning recommendations Share results with stakeholders (internal/external)	
		Thoughts	“Are these results strong enough to act on?” “Will this hold under real market conditions?” “Can I confidently defend this outcome?”	
		Pain Point	Risk of deviation between forecast and actual outcomes Pressure to justify results to stakeholders/regulators Translating complex outputs into clear decisions	
		Emotions	Satisfaction when results align with expectations Pressure before final submission or approval Confidence when outcomes are accepted and validated	
		Tool	PLEXOS final output reports Excel / BI dashboards for reporting ISO DAM systems and stakeholder reporting tools	